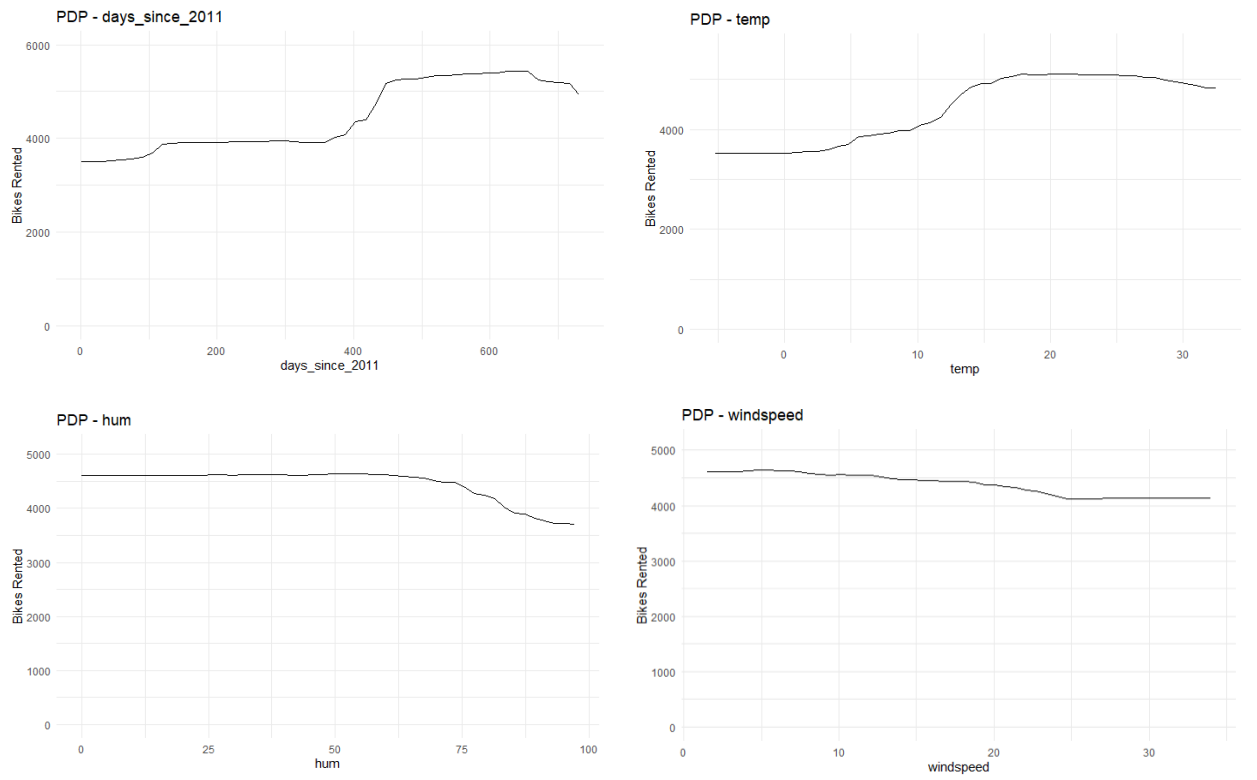


XAI: Model-agnostic methods

Exercise 5.- Model-agnostic: Partial Dependency Plot (PDP)

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1.- One dimensional Partial Dependence Plot



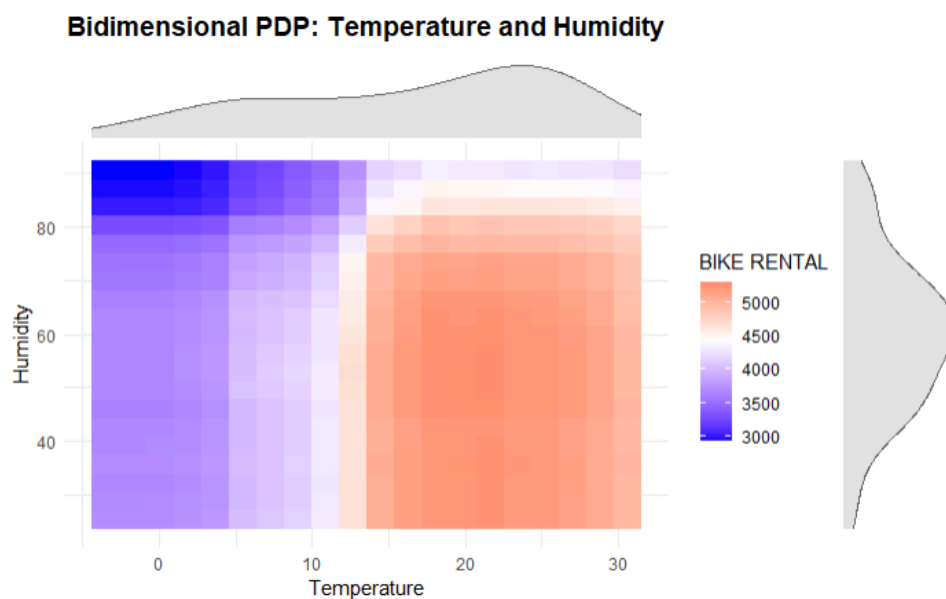
1- Partial Dependency Plot of the most Important features

Analysis of Partial Dependence Plots (PDP):

- The days_since_2011 variable has a positive effect on the predicted value of bikes rented, showing that the business is growing over time, with a slight fall in the last recorded days.
- When the temperature increases, the predicted value of rented bikes tends to increase as well, saying that with good weather more people use the bike. Still, if the temperature value is too high, growth stops, and it may even decrease the number of clients.
- The predicted value remains stable with humidity levels between 0 and 60, but as humidity approaches 100 it decreases.
- The speed of the wind seems to affect less to the predicted value for bike rental, though with a minor negative trend.

The model reveals that bike rental demand is primarily driven by long-term growth and thermal comfort. While there is a clear regular expansion of the business over time, environmental factors act as critical thresholds: moderate heat encourages usage, but extreme temperatures, high humidity (above 60%), and high wind speeds serve as factors that predictably lower demand.

2.- Bidimensional Partial Dependency Plot

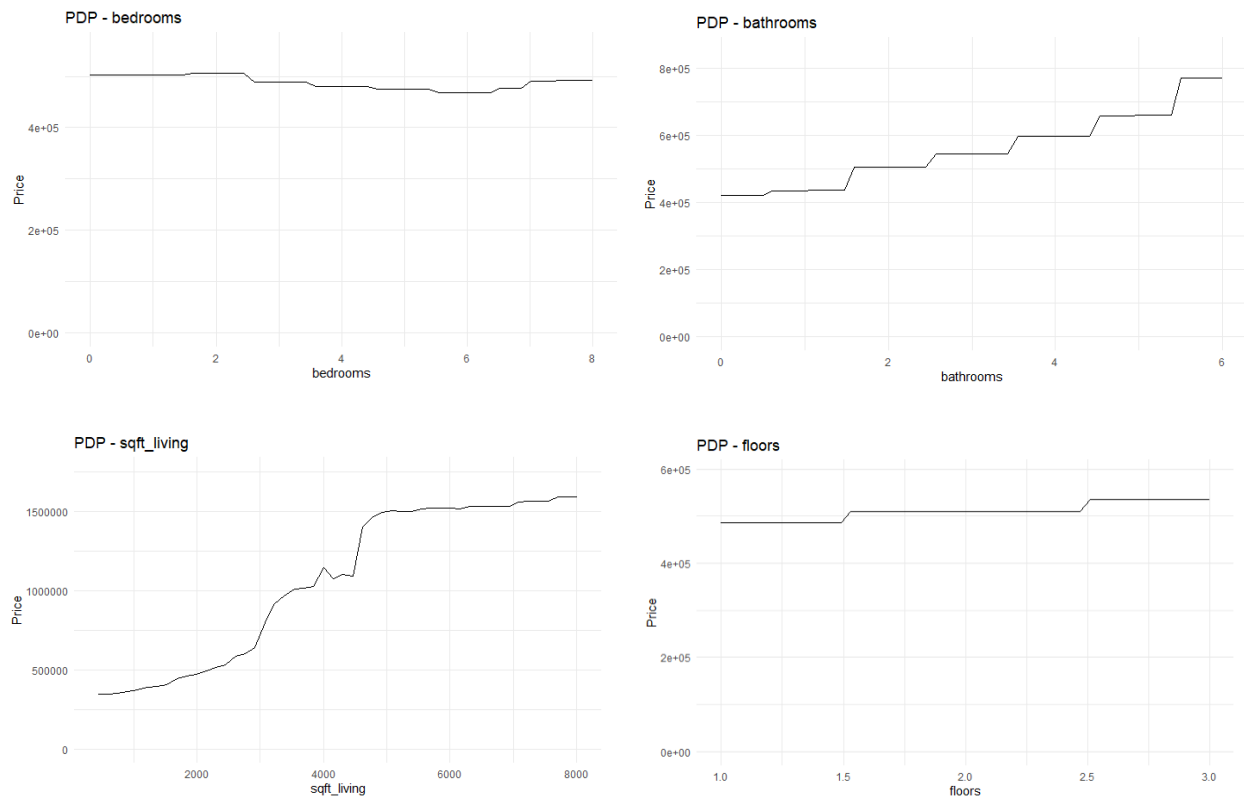


2- Bidimensional Partial Dependency Plot of Temperature and Humidity

The bidimensional plot shows a clear positive effect on the number of bikes rented when the temperature is high and the humidity is low. This effect is logical since these values usually mean good weather and more people choosing the bike over other vehicles such as the car or the bus.

On the other side of the plot, when temperature is low and humidity high, the number of bikes rented is at its lowest, which is totally understandable because using the bike with high humidity or rain is not the most pleasant (or most safe) way of moving around.

3.- PDP to explain the price of a house



3- Partial Dependency Plot of the most Important features

Analysis of Partial Dependence Plots (PDP):

- The number of bedrooms is not one of the main factors that affect the price. The final price of the house stays constant independently of the number of bedrooms.
- The number of bathrooms is one of the key variables affecting property value. As expected, there is a clear positive correlation. As the number of bathrooms increases, the house price rises accordingly, which is consistent with logical thinking.
- The size of the living room is the variable that has the most effect on the price, with the price increasing drastically when the square feet of the living room increase as well. Also the lowest prices are for houses with small living rooms, suggesting that a comfortable common space is one of the keys for customers when choosing a place to live.
- The number of floors is not a primary driver of property value. While it exhibits a slight positive correlation with price, its overall impact on the model's prediction is marginal.

In conclusion, living room square feet is the most critical predictor of house prices, followed by the number of bathrooms. In contrast, other variables such as floor count and number of bedrooms have almost no effect.